Math 534H - Spring 2026 - Lecture 9

Recall from last time we were reduced to:

$$(\dagger) \quad \boxed{u_t - c u_x = g(x-ct)}$$

Need to solve this to find u.

?

This is an inhomogeneous transport equation.

So we know that $\boxed{u(x,t)} =$ sol. to the homog. equation $+$ particular sol. of inhomog. problem.

Homogeneous equation

$u_t - c u_x = 0$ Charact lines $x + ct = \text{const}$.

So the homog. solution $\boxed{h(x+ct)}$ for any $h: \mathbb{R} \to \mathbb{R}$ differentiable fc. of one variable.

<u>Strauss</u>: Guesses that a solution to

$u_t^{inh} - c u_x^{inh} = g(x-ct)$ should look like a

function $\underline{u^{inh}(x,t) = f(x-ct)}$ (*) where f as a

function of <u>one variable</u> satisfies that

$f'(s) = \dfrac{g(s)}{-2c}$. <u>Why</u>? Because if (*)

is true then $\dfrac{\partial u^{inh}}{\partial t} = f' \cdot (-c)$

$$\frac{\partial u^{inh}}{\partial x} = f' \cdot 1$$

But $\underbrace{u_t^{inh}} - c u_x^{inh} = g \iff f'(-c) - c \cdot f' = g$

$$\iff -2c f' = g \implies \boxed{f' = \frac{g}{-2c}}$$

Hence the solution $u(x,t)$ of $(\dagger)$ is

$$u(x,t) = h(x+ct) + f(x-ct)$$

where h, f are differentiable functions of one variable $\mathbb{R} \to \mathbb{R}$ and f satisfies $f'(s) = \frac{-g(s)}{2c}$.

But then if go backward we get that in fact this $u(x,t)$ solves $\underline{\Box u = 0}$.

So all in all, the general solution to wave eq. $\boxed{\Box u = 0}$

$$\boxed{u(x,t) = h(x+ct) + f(x-ct)}$$

<u>where f, h are (twice) differentiable functions from $\mathbb{R} \to \mathbb{R}$.</u>

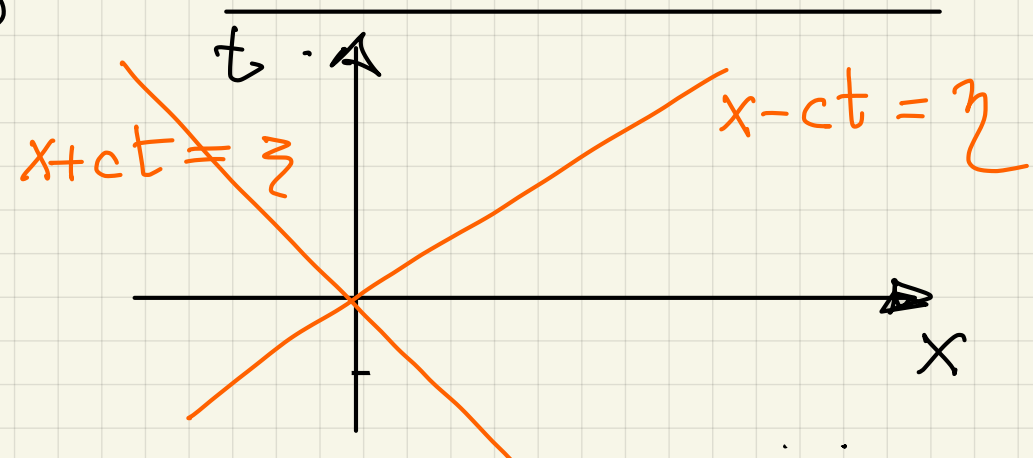
$\hookrightarrow C^2$

<u>Remark</u>: Going backward in how started g was itself a general differentiable function of one variable $g : \mathbb{R} \to \mathbb{R}$. So in fact f' and hence f is also a general differentiable function of one variable $f : \mathbb{R} \to \mathbb{R}$.

Want to revisit finding the general solution to $\Box u = 0$ in two different ways.

<u>Second Take</u>: using the <u>Coordinate Method</u>.

(a <u>change of variables</u>)



Let $\begin{cases} \xi = x + ct \\ \eta = x - ct \end{cases}$

Then let's find out what is $\Box = \partial_{tt} - c^2 \partial_{xx}$ in the new variables (ξ, η).

$$(\partial_t + c\partial_x)(\partial_t - c\partial_x)$$

$$u(x,t) \to u(\xi, \eta)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} \cdot 1 + \frac{\partial u}{\partial \eta} \cdot 1$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \xi} \cdot \frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \eta} \cdot \frac{\partial \eta}{\partial t} = \frac{\partial u}{\partial \xi} c + \frac{\partial u}{\partial \eta}(-c)$$

Then $$(\partial_t + c\partial_x) = c\left(\frac{\partial}{\partial \xi} \cancel{+ \frac{\partial}{\partial \eta}}\right) + c\left(\frac{\partial u}{\partial \xi} \cancel{+ \frac{\partial u}{\partial \eta}}\right)$$

$$= 2c \frac{\partial}{\partial \xi}$$

$$(\partial_t - c\partial_x) = c\left(\cancel{\frac{\partial}{\partial \xi}} - \frac{\partial}{\partial \eta}\right) - c\left(\cancel{\frac{\partial}{\partial \xi}} + \frac{\partial}{\partial \eta}\right)$$

$$= -2c \frac{\partial}{\partial \eta}.$$ So:

$$\Box_{\xi\eta} = -4c^2 \partial_\xi \partial_\eta$$

So solving $\Box u = 0 \iff -4c^2 \dfrac{\partial^2 u}{\partial\xi\partial\eta} = 0$ in the ξ, η variables

$$\iff \boxed{\partial_\xi \partial_\eta u = 0} \iff$$

$$\underline{u(\xi, \eta) = h(\xi) + f(\eta)}$$ where f and h are any differentiable functions $\mathbb{R} \to \mathbb{R}$.

Coming back to $(x,t) \iff$ to what we had before $\left(\begin{array}{l} \xi = x+ct \\ \eta = x-ct \end{array}\right) \to \underline{u(x,t) = h(x+ct) + f(x-ct)}$

<u>Third take</u> : Use the Duhamel integral as we did in Lecture 6 to solve the inhomogeneous transport equation: $u_t - c u_x = F(x,t)$

$F(x,t) = g(x-ct)$ later at very end.

Recall that sol. homog. equation is $h(x+ct)$, $h : \mathbb{R} \to \mathbb{R}$ any differentiable fc.

Here if we <u>repeat</u> what <u>we did in lecture 6</u>, parametrizing the <u>characteristic lines</u> which are $\underline{\chi(s) = (x - cs, t+s)}$ we get

$$\boxed{u(x,t) = h(x+ct) + \int_0^t F(\underbrace{x-(\tau-t)c}_{1^{st}\text{ variable}}, \underbrace{\tau}_{\text{second variable}})\, d\tau}$$

Now we recall $F(x,t) = g(x-ct)$ so the Duhamel integral becomes

$$\int_0^t g\big(\underbrace{x-(\tau-t)c-c\tau}_{x-2c\tau+ct}\big)\,d\tau =$$

$$y = x-2c\tau+ct$$
$$dy = -2c\,d\tau$$
$$d\tau = \frac{-1}{2c}\,dy$$

$$-\frac{1}{2c}\int_{x+ct}^{x-ct} g(y)\,dy$$

$$= \frac{1}{2c}\int_{x-ct}^{x+ct} g(y)\,dy.$$

All in all we get

$$\boxed{u(x,t) = h(x+ct) + \frac{1}{2c}\int_{x-ct}^{x+ct} g(y)\,dy} \quad (+)$$

and actually tracing back this $u(x,t)$ solves the wave equation $\boxed{\Box u = 0}$ and we can rewrite $(+)$ as

$$\boxed{u(x,t) = H(x+ct) + G(x-ct)}$$ for

any $H, G : \mathbb{R} \to \mathbb{R}$ (twice) differentiable.

Now suppose we are given the initial value problem (IVP)

$$\begin{cases} \Box u = 0 \\ u(x,0) = \phi(x) \\ u_t(x,0) = \psi(x) \end{cases}$$

Q: What's $u(x,t)$?

CLASS HMWK you read how Strauss does it using $u(x,t) = h(x+ct) + f(x-ct)$, $f, h : \mathbb{R} \to \mathbb{R}$.

Here we are going to use (†) at the top of the previous page

$$\phi(x) = u(x,0) = h(x) + 0 \Rightarrow \boxed{h = \phi}.$$

$$u_t(x,t) = c\,h'(x+ct) + \frac{1}{2c}\left[g(x+ct)\cdot c + c\,g(x-ct)\right]$$

(FTC).

$$u_t(x,t) = c\,h'(x+ct) + \frac{1}{2}\left[g(x+ct) - g(x-ct)\right]$$

Next plug $t=0$ and use $u_t(x,0) = \psi(x)$ to get

$$\psi(x) = c\,\phi'(x) + g(x) \rightarrow \boxed{g(x) = \psi(x) - c\,\phi'(x)}$$

All in all we get D'Alambert formula for the solution of the IVP for the wave eq.

$$\boxed{u(x,t) = \frac{1}{2}\Big[\phi(x+ct) + \phi(x-ct)\Big] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)\,dy}$$

<u>NEXT TIME</u> we do some examples and start <u>Section 2.2</u> on <u>Causality</u> (Finite prop. speed; domain of dependence; domain of influence) and <u>Energy for wave</u>